Ex. No: 14 File Allocation Strategies

PROGRAM A: SEQUENTIAL FILE ALLOCATION

C PROGRAM

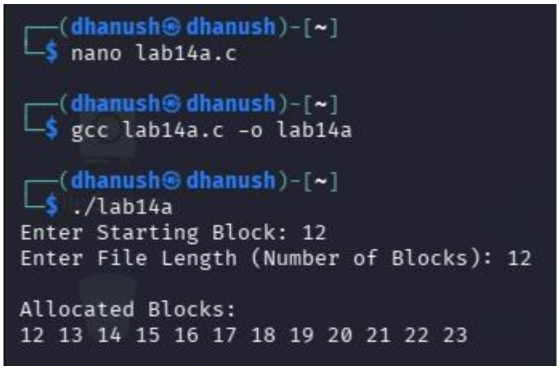
```c
#include <stdio.h>
int main()
{
int start, length, i;
printf("Enter Starting Block: ");
scanf("%d", &start);
printf("Enter File Length (Number of Blocks): ");
scanf("%d", &length);
printf("\nAllocated Blocks:\n");
for(i = 0; i < length; i++)
{
printf("%d ", start + i);
```

Ex. No: 14

File Allocation Strategies


```c
}
printf("\n");
return 0;
}
```

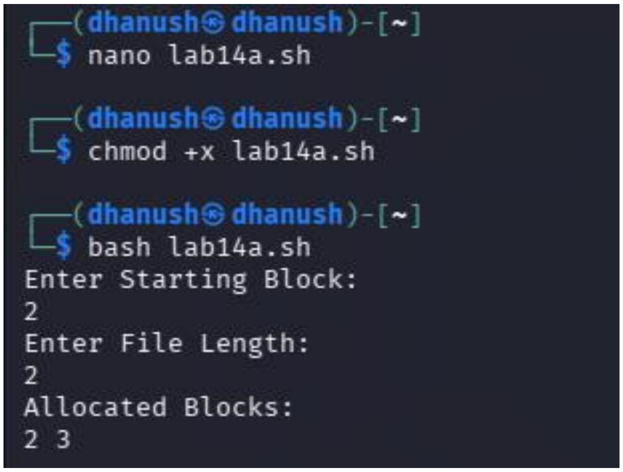
Output:



SHELL SCRIPT

```
#!/bin/bash

echo "Enter Starting Block:"

read start

echo "Enter File Length:"

read length

echo "Allocated Blocks:"

for ((i=0;i<length;i++))

do

echo -n "$((start+i)) "

done

echo
```

Output:



PROGRAM B: INDEXED FILE ALLOCATION

C PROGRAM

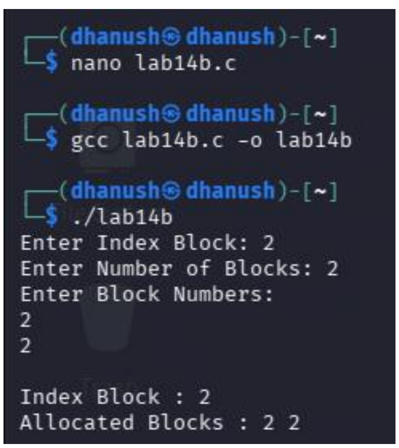
```
#include <stdio.h>

int main()

{

int n, indexBlock, blocks[20], i;

printf("Enter Index Block: ");

scanf("%d", &indexBlock);

printf("Enter Number of Blocks: ");

scanf("%d", &n);

printf("Enter Block Numbers:\n");
```



```
for(i = 0; i < n; i++)

{
```

```c
scanf("%d", &blocks[i]);

}

printf("\nIndex Block : %d\n", indexBlock);

printf("Allocated Blocks : ");

for(i = 0; i < n; i++)

{

printf("%d ", blocks[i]);

}

printf("\n");

return 0;

}
```

Output:



```
┌──(dhanush㉿dhanush)-[~]
└─$ nano lab14b.c

┌──(dhanush㉿dhanush)-[~]
└─$ gcc lab14b.c -o lab14b

┌──(dhanush㉿dhanush)-[~]
└─$ ./lab14b
Enter Index Block: 2
Enter Number of Blocks: 2
Enter Block Numbers:
2
2

Index Block : 2
Allocated Blocks : 2 2
```

SHELL SCRIPT

```bash
#!/bin/bash

echo "Enter Index Block:"

read index

echo "Enter Number of Blocks:"
```

```
read n
echo "Enter Block Numbers:"
for ((i=0;i<n;i++))
do
read block[$i]
done
echo "Index Block : $index"
echo -n "Allocated Blocks : "
for ((i=0;i<n;i++))
do
echo -n "${block[$i]} "
done
echo
```

Output:

```
┌──(dhanush㉿dhanush)-[~]
└─$ nano lab14b.sh

┌──(dhanush㉿dhanush)-[~]
└─$ chmod +x lab14b.sh

┌──(dhanush㉿dhanush)-[~]
└─$ bash lab14b.sh
Enter Index Block:
2
Enter Number of Blocks:
2
Enter Block Numbers:
2

Index Block : 2
Allocated Blocks : 2
```